

[Joyn.de] Test plan

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Introduction

Joyn is a German streaming app to enjoy television at any time and free of charge as web TV. Joyn offers a huge selection to watch over 60 live TV channels via the Internet. In order to use app features, users have to register. A watch list helps to get recommendations that are tailored to a user. In order to achieve full entertainment. Joyn is developed as a hybrid application so it can be used not only as a Web app but also as a mobile app. However, Joyn is not only television in the live stream but also offers many programs after TV broadcast. The search feature allows users to find the specific show by writing their name in the search field. In order to watch everything in HD, all originals and also much more complete series and TV formats before they are broadcast on TV users can buy Joyn PLUS+.

1. Test strategy

1.1 Scope of testing

1.1.1 Feature to be tested (web)

Module name	Login type	Description
Registration	User logged out	Users can go to the registration form by clicking on the Login overlay. Registration opens with the following parts: - Joyn logo - headline "Registrierung" - 1 textbox for email and next open 1 textbox for password - 1 textbox for the birth date - 3 checkboxes for gender: "weiblich", "männlich", "divers" - clickable text with terms of condition (AGB) and data security (Datenschutzbestimmungen) marked bold, will open in the new tab. - "Kostenlos registrieren" button (enabled as soon as both text boxes are filled with valid data) - SSO logins for: Facebook, Google, Apple, and Amazon Valid data = email with "@" and "." and <= 60 chars; password >= 6 Focused textboxes are marked with white line Clicking the registration button gets you back to the home page. There is a colored circle around the account icon in the top right.

Log in	User logged out	Users can be accessed via the log-in link on the top right of the header. The login page contains: - Joyn logo - headline "Login" - 1 textbox for email and next open 1 textbox for password - clickable "Passwort vergessen?" forgot password string - Login button (enabled as soon as both textboxes are filled) Valid data = email with "@" and "." and =< 6 chars; password >= 6 There is a colored circle around the account icon in the top right.
Searching feature	The user logged out/ logged in	Users can search by clicking on the magnifying glass button top right in the header. Entering any invalid term -> a "no results" message is displayed beyond the search bar, and no assets are displayed, Entering any valid term-> search results are displayed in a listview beyond the search bar. Adding/Removing letters to the search triggers search results to update immediately. Search terms are stored when navigating to a detail page and then using the back button.
Ad Blocker	User logged out/ logged in	Users can turn on an adblocker of their choice, Free content with ads (everything that is not marked with a PLUS) should not be playable with an active AdBlocker. There should be an Error message: Hoppla! Da ist etwas schief gelaufen.
Bookmarks & Bookmarks lane	User logged in	The bookmark lane can contain a maximum of 20 asset cards. After that, an "Alle >" link is displayed above the lane. If the user clicks on the link, a grid view is opened containing. Users can see a bookmark icon on every detail page (series, movie, and compilations) and on all standard cards. If the user taps on the bookmark icon: - Logged out user: the login screen appears -> after login -> the bookmark icon is filled - Logged-in user: the icon on the button is filled. As soon as at least 1 asset is bookmarked the bookmarks lane (header title "Merkliste") is displayed on home with the latest bookmarked asset in 1st position. If a user taps on a filled bookmark icon, the icon is unmarked and the asset card is removed from the bookmark lane. HINT: Bookmarks should also be shown when you log in with the same account on a different device/browser.

Cards: Channel Card	User logged out/ logged in	Users can see it in Search or in Channel lane on HOME (called Mediathek). Channel cards show - the channel logo - the vibrant color of the channel label (on the Desktop shown on the hover) Clicking on a channel card opens a grid view, with the channel name as the headline. Channel cards are only available for channels with VOD content - no empty grid views.
Cards: Feature Cards	User logged out/ logged in	User can see on HOME screen - currently named "Originals & Exclusives". <u>Details:</u> This lane has featured content. Cards here contain: - Art logo/headline - For desktop on hover the genre, vibrant color, and the short description appears - Clicking on any asset card leads to the asset detail page.
Cards: Standard Cards	User logged out/ logged in	User can find grid view, in lanes, on HOME, in search. Standard cards: • cover image • brand logo in the top right • an art logo /headline • vibrant color appears on hover • For a desktop there is a hover effect which shows a short description (not on search overlay), a "Mehr" button and a bookmark button • Clicking an asset card leads to a detailed page Series cards: all of the above for Standard Cards, plus a number of available seasons Movie cards: all of the above for Standard Cards, plus genre
Collection lane	User logged out/ logged in	User can see Lane on start page e.g. "Themen für dich". Cards have the same width as a standard card but a smaller height, they contain:

		- Picture - Name - gradient on the bottom - Vibrant color (on hover for Desktop, always shown on mobile).
Detail Page: Collection	User logged out/ logged in	User can see Detail Page clicking on a card on the collection lane. Collection detail page contains: - Headline - Description - Background image - Content cards in grid view, or lane view - Collection lane Hover effect on the cards has the same behavior as standard lanes. Clicking on a card leads to the detail page of that asset.
Playback: Live TV (UI)	User logged out/ logged in	User can found in the main navigation. The Live TV UI contains: • Vibrant color gradient in the background. (should be the same as channel vibrant color) • Close icon. • Progress bar. • Fullscreen icon. • Sound volume icon. • "Programm" button = show EPG (electronic program guide). Note that there's no button for mobile in portrait mode, instead the EPG is displayed below the video.
Detail Page: Movie	User logged out/ logged in	User can click on a movie card the movie detail page opens. <u>Elements:</u> • cover • art logo • short description • more metadata like FSK (on more-overlay), channel logos, duration • "Mehr" button • play/resume (with progress bar) button labeled with "Jetzt ansehen" or "Jetzt weiterschauen" • bookmark button • recommendation lane

		<u>Details:</u> - Clicking the "Mehr" button opens the overlay, also displaying all metadata and short description. Can be dismissed via a close button or clicked somewhere in the overlay. - Clicking the play/resume button starts/resumes the asset.
Detail Page: Series	User logged in	User can see this page on HOME or in search with results click on any series asset card. When clicking on an asset card the series detail page opens. <u>Elements:</u> • cover • art logo • number of available seasons • short description • more metadata like FSK (in more overlay), channel logos • "Mehr" button • play/resume (with progress bar) button labeled with newest/latest watched episode • bookmark button • season selector • episode selector • recommendations lane • cards may contain progress bars (if watched before) <u>Details:</u> • Clicking "Merh" button opens the overlay, also displaying all metadata and a short description. Can be dismissed via a close button or clicking anywhere • Clicking the play/resume button starts/resumes episode labeled on the button. • If the content is also available with PLUS, an extra lane appears. (e.g. series "Check Check.") - clicking on an episode opens the upselling overlay.
Live Lane	User logged out/ logged in	User can find on HOME beyond the featured lane - possibly named "LIVE TV". This lane shows currently running Live TV content. Cards here contain: - a nearly full-width image

		- a LIVE tag - Airtimes - Play button + program title There is NO "All" button. Clicking on any asset card switches to the LIVE TV section, starting the currently selected program.
Sports: Competition Page	User logged out/ logged in	On the sports page, there are different sports lanes. This button leads to a competition page. It contains: - Headline - Description - Background image - Different lanes, e.g. Highlight lane, lanes for every day of the event - Sports cards on the lane behave like sports cards on sport page

Feature to be tested (mobile)

Module name	Login type	Description
Ads	User logged out/ logged in	User sees advertisements on most videos without PLUS marker Ad UI contains: - Number of ads - Link ("Zum Werbetreibenden") => clicking opens the advertiser page in a new tab; ad is paused - Total time countdown Ad must not be skippable: - If you put it in the background -> ad should pause - If you minimize -> ad should pause - No possibility to fast forward in the ad Hover effect - when moving the mouse pointer or touching the display: - The pause button & 'X' button are shown
Bookmarks &	User logged out/ logged in	User can see a bookmark icon on every detail page (series, movie, and compilations) and on all standard cards. If user taps on the

Bookmarks lane		bookmark icon: - Logged out user: the login screen appears -> after login -> the bookmark icon is filled - Logged in user: the icon on the button is filled. As soon as at least 1 asset is bookmarked the bookmarks lane (header title "Merkliste") is displayed on home with the latest bookmarked asset in 1st position. The bookmark lane can contain a maximum of 20 asset cards. After that, an "Alle >" link is displayed above the lane. If the user clicks on the link, a grid view is opened containing all bookmarked assets. If a user taps on a filled bookmark icon, the icon is unmarked and the asset card is removed from the bookmark lane. HINT: Bookmarks should also be shown when you log in with the same account on a different device/browser.
Cards: Channel Card	User logged out/ logged in	User can see in Search or in Channel lane on HOME (called Mediathek). Channel cards show - the channel logo - the vibrant color of the channel label Clicking on a channel card opens a grid view, with the channel name as the headline. Channel cards are only available for channels with VOD content - no empty grid views.
Cards: Feature Cards	User logged out/ logged in	User can see on HOME screen - currently named "Originals & Exclusives". <u>Details:</u> This lane has featured content. Cards here contain: - A nearly full-width image on mobile - Art logo/headline - Short description on mobile - Clicking on any asset card leads to the asset detail screen.
Cards: Standard Cards	User logged out/ logged in	User can find grid view, in lanes, on HOME, in search. Standard cards: • cover image • brand logo in the top right • an art logo /headline • vibrant color on mobile, for desktop it appears on hover

		- For desktop there is a hover effect which shows a short description (not on search overlay), a "Mehr" button and a bookmark button - Clicking an asset cards leads to a detail page Series cards: all of the above for Standard Cards, plus number of available seasons Movie cards: all of the above for Standard Cards, plus genre Compilation cards: all of the above for Standard Cards, plus amount of videos
Collection lane	User logged out/ logged in	User can see Lane on start page e.g. "Themen für dich". Cards have the same width as a standard card but a smaller height, they contain: - Picture - Name - gradient on the bottom - Vibrant color (always shown on mobile).
Detail Page: Collection	User logged out/ logged in	User can see Detail Page clicking on a card on the collection lane. Collection detail page contains: - Headline - Description - Background image - Content cards in grid view, or lane view - Collection lane Hover effect on the cards has the same behavior as standard lanes. Clicking on a card leads to the detail page of that asset.
Detail Page: Series	User logged out/ logged in	User can see this page on HOME or in search with results click on any series asset card. When tapping on an asset card the series detail page opens. <u>Elements:</u> - cover - art logo - number of available seasons - short description - more metadata like FSK (in more-overlay), channel logos "Mehr" button

		• play/resume (with progress bar) button labeled with newest/latest watched episode • bookmark button • season selector • episode selector • recommendations lane • cards may contain progress bars (if watched before) <u>Details:</u> • Tapping "more" button opens the overlay, also displaying all metadata and short description. Can be dismissed via a close button or clicking anywhere • Tapping the play/resume button starts/resumes episode labeled on the button. • If the content is also available with PLUS, an extra lane appears. (e.g. series "Check Check.") - clicking on an episode opens the upselling overlay.
Live Lane	User logged out/ logged in	User can find on HOME beyond featured lane - possibly named "LIVE TV". This lane shows currently running Live TV content. Cards here contain: - a nearly full-width image - a LIVE tag - Airtimes - Play button + program title There is NO "All" button. Clicking on any asset card switches to the LIVE TV section, starting the currently selected program.
Playback: Live TV (UI)	User logged in	User can find this feature in the main navigation. Playback: Live TV (UI) should contain: • Vibrant color gradient in the background. (should be the same as channel vibrant color) • Close icon. • Progress bar. • Remaining time of the current asset. • Fullscreen icon.

		• Sound volume icon. • "Programm" button = show EPG (electronic program guide). Note that there's no button for mobile in portrait mode, instead the EPG is displayed below the video.
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1.1.2 Feature not to be tested

- Website's security
- Users' databases
- App performance

1.2 Test type

There are 3 types of testing for the Joyn project:

- Smoke testing
- Sanity testing
- Regression testing

1.3 Risk and issues

Risk	Mitigation
Network connection issues	Replace disconnected teammate with another, reassign responsibilities
Lack of motivation	Encourage team members with the moral support
Power outage in workplaces	Install a spare power generator
Environment malfunctions	Fix mechanical issues or replace with another device
Team member sickness	Replacement duties management

1.4 Test logistics

1.4.1 Who will test?

Web feature testing - Oksana Martyniuk, Oleg Lukyanchuk

Mobile feature testing - Roman Labonin, Ilya Kosterev

Documentation testing - Kate Matviichuk
Permission testing - Danylo Kotliarenko

1.4.2 When will the test occur?

- Test environment set up
- Decomposition created
- RTM is completed with test cases
- Test data and test users are ready

2. Test objective

The test objectives are to verify the functionality of the website Joyn.de. The project should focus on users' interaction with the website. The user should be able to use the functionality of the site to watch their favorite TV shows, news, and movies. The operation of the site must be continuous and uninterrupted. The user must have access to the functions in full or in part if he is not logged in.

3. Test criteria

3.1 Suspension criteria

If there are more than 40% failed test cases testing process is stopped

3.2 Exit criteria

The test process will be stopped at the end of the current sprint. Critical functionality is covered by test cases and they are 100% executed and passed with a coefficient of 90%.

4. Resource planning

4.1 System resource

No.	Resources	Description
1.	Server	Need Apache server, MySQL server, NGINX server, APN server.

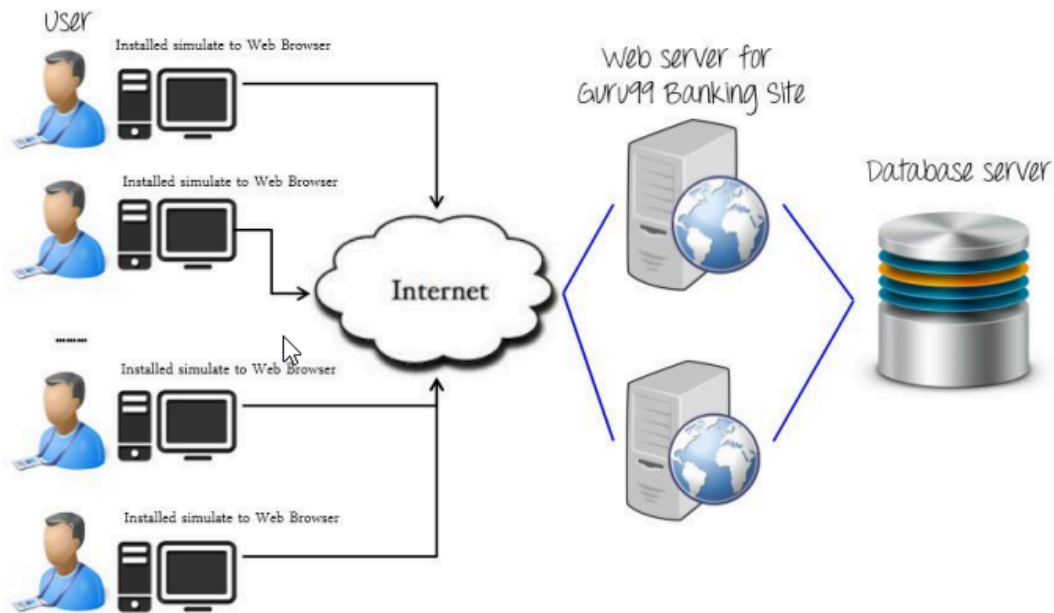
2.	Jira / TestRail	To track bug reports and last cases.
3.	Computer/browser	Desktop or laptop (macOS) with Windows 10 / Google Chrome
4.	Mobile phone	Phone with Android 11, Android 8.1.0
4.	Network	Internet line with a speed of at least 5 MB/s
5.	VPN	Connect to Germany IP through VPN-free.pro (for web)
6.	Screenshots / Video capture Tools	Loom (for web) / Bandicam (for web), Screen Recorder (for mobile)
7.	AdBlocker	AdBlock version 4.46.1

4.2 Human resources

No.	Resources	Description
1.	Mentor	Assigns task performers among students, monitors the implementation of the testing schedule, and checks the implementation of the project.
2.	Scrum master	Create integration Jira and TestRail.
3.	QA Students	Create test infrastructure, test cases and execute them, and log results. Report bugs and enter them into the bug tracking system. Create Test Result Report.

5. Test environment

The Test Environment should be set up as figure below:



6. Schedule & estimation

6.1 All project tasks and estimation

Task	Members	Estimate effort
Create Test-plan	QA Students	10 man-hour
Create decomposition for web and mobile.	QA Students	10 man-hour
Create integration Jira and TestRail.	Scrum master	4 man-hour
Create permission testing	QA Students	6 man-hour
Create Requirement Traceability Matrix	QA Students	8 man-hour
Create Test Cases	QA Students	16 man-hour
Execute test cases (Web), log results, enter bugs at Jira	QA Students	10 man-hour
Execute test cases (Mobile), log	QA Students	10 man-hour

results, and enter bugs at Jira		
Create a Test Result Report	QA Students	4 man-hour
Check up on the project	Mentor	2 man-hour
Total:		80 man-hours

6.2 Schedule to complete these tasks

Task	Start date	End date
Create Test-plan	10.05.2022	11.05.2022
Create decomposition for web and mobile.	10.05.2022	11.05.2022
Create integration Jira and TestRail.	10.05.2022	10.05.2022
Create permission testing	11.05.2022	11.05.2022
Create Requirement Traceability Matrix	12.05.2022	12.05.2022
Create Test Cases	11.05.2022	12.05.2022
Execute test cases (Web), log results, and enter bugs at Jira	12.05.2022	13.05.2022
Execute test cases (Mobile), log results, and enter bugs at Jira	12.05.2022	13.05.2022
Create a Test Result Report	16.05.2022	16.05.2022
Check up on the project	16.05.2022	16.05.2022

7. Test deliverables

7.1 Before testing phase

- Decomposition

- Test Cases documents

7.2 During the testing

- Requirement Traceability Matrix
- Test Data
- Bug reports

7.3 After the testing cycle is over

- Test Result Report